PANELSYNC

Full Product Scope — v2.0

PanelSync is a creative operating system built specifically for comic makers. It is the first tool to bring script writing, world building, character and location libraries, visual references, timelines, and story planning into a single coherent workspace — purpose-built for the way comics are actually made.

Core Philosophy	*The depth is always there for the people who want it. It never gets in the way of the people who don't. A writer can open PanelSync, create a World, and start writing without ever touching the Timeline, the Chalkboard, or the World Bible. Everything populates quietly in the background.*

1. Product Vision

PanelSync exists because comic creators are currently stitching together a workflow from tools that were never built for them. Scripts live in Google Docs. Character references live in Apple Notes or scattered folders. Timelines live in spreadsheets. World-building lore lives nowhere in particular. Nothing talks to anything else.

The frustration is not just organizational. It is creative. When you cannot find what you wrote about a character three months ago, you lose the thread. When your collaborator does not know what you decided about a location, you lose continuity. When you cannot see where a scene fits in the larger story, you lose perspective.

PanelSync solves all of this in one place, purpose-built for comics, with an interface that feels as natural on an iPad with an Apple Pencil as it does on a desktop with a keyboard.

1.1 Target Users

- Solo writer-artists who want one place to do everything
- Writer-artist pairs who need to stay organized and aligned across a shared project

- Small studio teams with editors, multiple writers, and multiple artists working in a shared universe
- Publishers and editors who need a 2000ft view across multiple series simultaneously

1.2 The Core Insight

PanelSync is not primarily a script editor with features bolted on. It is a world-first creative workspace. The World — the fictional universe the story inhabits — is the organizing principle. Everything else lives inside and refers back to the World.

2. Information Hierarchy

Structure	*Worlds Dashboard → World → Series → Issues → Pages*

- Worlds Dashboard — home screen. Shows all Worlds the user owns or is a member of. Entry point for creating or entering a World.
- World — a fictional universe. Contains the World Bible, Timeline collection, Chalkboard collection, Relationship Map, Mood Board, Notes, and all Series. World-level content is shared across every Series set in that World.
- Series — a single comic series within the World. Has series-specific arc notes, character and location overlays, and a series-level timeline. Contains all Issues.
- Issue — a single issue. Contains its Script and Storyboard pages, issue notes, and status tracking.
- Page — a single comic page. Always has a paired script page and storyboard canvas.

2.1 What Lives Where

Level	Content
World	World Bible (Characters, Locations, Factions, Objects, Lore), Timeline collection, Chalkboard collection, Relationship Map, Mood Board, Notes, Collaboration settings
Series	Series arc notes, series-specific character/location overlays, series timeline, series Chalkboard canvases

Issue	Script, Storyboard, issue notes, page status tracking, Pacing View, Script Breakdown grid
Page	Script page (comic-native formatted text) paired with Storyboard page (drawable canvas, reference images, sketch layer)

2.2 Progressive Disclosure

The sidebar defaults to showing only what the user needs — their series, issues, and pages. The World Bible, Timeline, Chalkboard, and other tools are accessible but not pushed. A writer who only wants to write scripts will never be confronted with tools they do not need.

3. Navigation & Sidebar

3.1 Worlds Dashboard

- Home screen when the app is opened
- All Worlds shown as cards with cover image, name, and last edited date
- Create a new World from the dashboard

3.2 Contextual Sidebar

- Top section — World-level tools: World Bible, Timeline, Chalkboard, Relationship Map, Mood Board, Notes
- Middle section — Series list, expandable to Issues, expandable to Pages
- Bottom section — current page context: quick access to the script and storyboard for the active page
- Collapsible on both iPad and web to maximize working space
- World switcher at top — jump to another World without returning to the dashboard

3.3 Split View

Any two sections can be viewed side by side. Common combinations are built in as quick-launch pairs:

- Script + Storyboard
- Script + Characters
- Timeline + Script
- Characters + Sketch canvas

- Locations + Sketch canvas
- Timeline + Characters

3.4 The 2000ft View

For editors and publishers, the sidebar can expand to a full modular dashboard showing multiple series timelines side by side, series status at a glance, and activity across the whole World. This is a power-user mode that does not affect the default experience.

4. Script Editor

The script editor is a first-class writing tool built specifically for comics — not a general text editor with comic formatting added on top.

4.1 Comic-Native Block Types

- Panel Block — auto-numbered. Cmd/Ctrl + Enter to insert.
- Scene Heading — INT./EXT., location, time of day. Recognized keywords: INT., EXT., WIDE, CLOSEUP, MEDIUM, HORIZON, BIRD'S EYE, WORM'S EYE.
- Description — action lines. Inline panel sizing tags: [page-width], [half], [1/3], [remainder].
- Dialogue — character name with World Bible autocomplete, dialogue text.
- Caption — narration or caption box text.
- SFX — sound effect, rendered large and stylized in the storyboard preview.

4.2 Editor Modes

- Standard — split view with script and storyboard preview side by side
- Distraction-free — full screen, everything hidden. Cmd/Ctrl + Shift + F.
- Outline — all content collapsed to panel headings only

4.3 Behaviour

- Tab cycles through block types within a panel
- Character name autocomplete from the series character registry
- Auto-numbering — panels renumber when one is added or deleted
- Auto-save — debounced 1.5 seconds after last keystroke

- Every script page is permanently paired with a storyboard canvas — adding a page in the script automatically creates a corresponding storyboard canvas

4.4 Script Export

- Industry-standard formatted comic script PDF for any issue
- Background job — user receives a download notification when ready

5. Storyboard

Every script page has a permanently paired storyboard canvas. In MVP, canvases are blank — the artist defines the layout by drawing. Auto-generation from the script is a v2 feature, but the data model and page structure are designed for it from day one.

5.1 Page Structure

- One storyboard canvas per script page, always paired, always in sync
- Full-page drawable surface — artist defines the panel layout by drawing it
- Panel grid guides available as a toggleable overlay
- Script for the current page accessible as a collapsible reference panel while drawing

5.2 Drawing Tools (iPad)

- Pencil tool with Apple Pencil pressure sensitivity
- Eraser, stroke weight slider, stroke color (black, dark grey, red for annotation)
- Two-finger tap: undo. Three-finger tap: redo.
- Pinch to zoom. Double-tap a panel area to enter focused mode.
- Toolbar auto-hides after 3 seconds; tap to reveal
- Page strip at bottom for thumbnail navigation

5.3 Reference Images

- Import any image from the device as a reference layer on any storyboard page
- Set reference images to reduced opacity for tracing
- Sketch layer always sits above the reference layer

5.4 Storyboard Views

- Script breakdown grid — all pages as thumbnails with status indicators
- Reading mode — sequential comic reader, full screen
- Presentation mode — pitch deck style walkthrough, script hidden
- Print layout view — shows bleed and margin guides

5.5 Auto-Generation (v2)

The script parser will derive panel layout, scene labels, speech bubble placeholders, and character proxy markers from the script and auto-populate the storyboard canvas. The page structure in MVP is designed for this.

6. World Bible

The canonical reference library for everything that exists in the World. Shared across all Series. Every entry has a profile, sketch/reference image layer, and timeline tags.

6.1 Entry Types

- Characters — name, role, appearance, backstory, relationships, series appearances
- Locations — name, description, time period, appearances, how it changes over time
- Factions — organizations and groups characters belong to
- Objects / Artifacts — significant items in the story
- Lore / Concepts — world rules, history, mythology, systems of magic or technology

6.2 Entry Structure

- Profile fields specific to the entry type
- Rich text notes — free-form writing
- Reference images — import images for visual reference
- Sketch layer — draw directly with Apple Pencil; trace over reference images
- Timeline tags — tag the entry or specific assets to a point or range on any timeline
- Series overlays — series-specific notes layer on top of the world-level entry without overwriting it
- Issue-specific notes — attached to a specific issue's use of an entry, searchable world-wide

6.3 Character Knowledge Tracking

Characters default to knowing everything relevant to the story. Writers tag knowledge gaps when narratively significant. These tags are stored on the character entry and on relevant timeline events, setting up the v2 Continuity Tracker.

6.4 Navigation

- Browse by entry type
- Global search — find any entry, note, or tagged asset from a single search box
- Filter by series or by timeline range

7. Timeline

One of PanelSync's most distinctive features — and the one no other comic tool provides. Writers create as many timelines as they need, each a named ordered record of events at any scale.

7.1 Timeline Intent Tags

All timelines share the same structure. The user sets an intent tag when creating a timeline:

- Story timeline — in-world events in the order they happen in the fiction
- Reference timeline — real-world or external events used as anchors
- Character arc timeline — scoped to a single character's journey
- Production timeline — real-world deadlines and milestones

7.2 Events

- Title, date label (free text), description, character tags, location tag, point or range marker
- Point tags — this happened at this moment
- Range tags — this state of affairs persisted from here to here
- Chronological order by default; manual drag-to-reorder available
- Every asset in the app — sketches, notes, script pages, World Bible entries — can be tagged to any event or range on any timeline

7.3 Named Ranges

- Group any span of events into a named range — 'Act One', 'The Civil War Years', 'Week of the Collapse'
- Named ranges are jumpable instantly from a range index
- Ranges can overlap — an event can belong to multiple named ranges

7.4 Navigation & Display

- Zoom — scale from centuries to hours
- Density toggle — compact (titles only) or expanded (full event details)
- Jump to character — filter to events involving a specific character
- Jump to date or era — type to jump instantly
- Jump to named range — select from the range index
- Color coding — by character, location, or event type
- Pin any timeline alongside any other for side-by-side comparison

7.5 Timeline Lenses (v2)

Reader knowledge, character knowledge, story order, and real-world reference lenses. The data model is designed for these from day one.

8. The Chalkboard

A freeform ideation canvas for story planning, character exploration, and breaking story structure. Where writers think out loud. Everything is non-destructive — ideas are never deleted, only promoted or abandoned.

8.1 Core Concept

- Create as many canvases as needed, scoped to any level — World, Series, or Issue
- Infinite zoomable surface of nodes and connections
- Non-destructive — abandoned paths remain visible as a record of the creative process

8.2 Node Types

- Plot event — something that happens
- Character moment — what a character does, feels, or decides
- Question — 'what if X?' or 'why does Y?'
- Cliffhanger — a moment of unresolved tension
- Branch point — where the story could go multiple ways
- Dead end — a path that was explored and abandoned

8.3 Promoting to Canon

- Lasso any set of nodes to select a path
- 'Promote to Canon' copies those nodes to the World Timeline — originals stay on the canvas
- Named alternate storylines — save any branch as a named alternate

8.4 Keyboard Input Mode

- Cmd/Ctrl + Right Arrow — new node to the right (continue the path)
- Cmd/Ctrl + Up Arrow — new node above (parallel thread)
- Cmd/Ctrl + Left Arrow — go back to the previous node
- Cmd/Ctrl + Up + Right Arrow — create a fork from the current node
- Cmd/Ctrl + Right + Up/Down Arrow — navigate up or down a fork
- Optional swim lanes and columns for writers who want imposed structure

8.5 Apple Pencil Card Mode (iPad)

- Each card fills the iPad screen — write or draw with the Pencil like a sharpie on a notecard
- Card surface is blank by default; dot grid or ruled background toggleable
- Three-finger swipe right — new card to the right
- Three-finger swipe up — new card above (parallel thread)
- Three-finger swipe down — new card below (alternate branch)
- Three-finger swipe left — go back to the previous card
- Handwriting preserved as-is by default — looks like your actual handwriting
- Handwriting recognition runs in background by default — cards are searchable unless recognition is toggled off in settings
- Display toggle switches between showing actual handwriting and the recognized text interpretation
- Mixed canvases supported — some cards typed, some handwritten

8.6 Canvas Organization

- Zoom levels — titles only when zoomed out, full content when zoomed in

- Path highlight mode — click any node to illuminate all connected nodes
- Color coding by node type
- Named alternate storylines browsable as branches

8.7 Collaboration

- Shared canvas — live presence, collaborators' cursors and edits visible in real time
- Private personal canvas per collaborator per project — invisible to others unless shared

9. Relationship Map

A living dynamic diagram of how characters in the World relate to each other. Updates as characters are added and relationships are tagged.

- Characters as nodes; relationships as labeled edges — alliance, conflict, family, romance, rivalry, mentor, secret
- Relationship state tags to timeline ranges — the map can show the state of relationships at any point in the story
- Filter by character or by relationship type
- Dashed lines indicate relationships one character does not know about

10. Mood / Tone Board

A curated visual reference library per Series or Arc. Less structured than the World Bible — more impressionistic, more visual. The tool for keeping writer and artist aligned on the feel of something.

- Import images, color swatches, and text snippets onto a freeform board
- Organize by mood, arc, character, or location
- Share with collaborators as a visual brief
- Reference images can be pulled into any World Bible entry or storyboard page

11. Pacing View

Shows the emotional beat structure of an issue or arc as a visual curve. For checking whether the story is structurally balanced.

- Each page tagged with a beat type — action, quiet, revelation, cliffhanger, humor, grief
- Rendered as a curve across the issue
- Compare curves across issues to check for consistent rhythm
- Click any point on the curve to jump to that script page

12. Continuity Tracker (v2)

Uses knowledge tags, physical state tags, and relationship state tags built into the MVP data model to automatically detect inconsistencies across the script.

- Knowledge inconsistencies — a character appears to know something before it was revealed to them
- Physical state inconsistencies — a character appears in a scene after being shown injured, dead, or elsewhere
- Relationship inconsistencies — a character acts on an alliance or conflict before it was established

Surfaced as inline flags on the script page and in a dedicated continuity panel. Flags can be toggled off per instance when the inconsistency is intentional.

13. Collaboration

13.1 Access Model

- Open by default — everyone with access can see and edit everything
- Individual sections lockable by the World owner
- Private notes and canvases — visible only to their creator

13.2 Roles

- Owner — full access, manages members, locks sections, can delete the World
- Editor — can edit everything that is not locked
- Commenter — comments and annotations only
- Viewer — read-only

13.3 Presence & Awareness

- Live presence indicator — who is active and where
- Named cursors on shared surfaces
- Page status tracking — draft, in review, locked, complete
- Read receipts — know when something you wrote has been seen

13.4 Comments & Annotations

- Threaded text comments on any script line, panel, or World Bible entry
- Drawn annotation marks on storyboard pages
- Voice note comments
- Comments resolved or dismissed by the page owner

13.5 Sync

- MVP — async sync. Last-write-wins with a visible 'last edited by' indicator.
- v2 — real-time collaborative sync using Yjs CRDT.

14. Export & Sharing

14.1 Exports

- Script PDF — industry-standard formatted comic script
- Series Bible PDF — characters, locations, lore, and timeline in one document
- Storyboard PDF — full pages at 300dpi
- Storyboard image sequence — each page as an individual PNG
- Chalkboard export — high-resolution image or PDF
- Relationship Map export — diagram as an image

14.2 Read-Only Sharing

- Read-only live link to any section — script, storyboard, World Bible entry, timeline, Chalkboard canvas
- Links can be password-protected or set to expire

- No PanelSync account required to view

15. Notes

- Create notes at any level — World, Series, Issue, or attached to a specific character, location, or timeline event
- Folder organization with drag-and-drop
- Rich text editor — bold, italic, headings, bullet lists, inline code
- Quick-capture — global shortcut opens a floating note from anywhere in the app
- Full-text search across all notes in a World
- Private notes visible only to their creator

16. Platform

16.1 The Unified Approach

One Expo (React Native) codebase runs natively on iPad and in the browser via Expo for Web. Shared components cover everything — script editor, notes, timeline, navigation, design system. Platform-specific code is isolated to the drawing canvas only.

16.2 Platform Feature Comparison

Feature	iPad	Web
Script editor — full comic-native formatting	Full	Full
Storyboard — drawable canvas	Full (Skia + Apple Pencil)	Read-only preview v1
Drawing on character/location pages	Full	Not available v1
Reference image import + trace	Full	Import only v1
Timeline — create, edit, navigate	Full	Full
Chalkboard — keyboard mode	Full (hardware keyboard)	Full
Chalkboard — Apple Pencil card mode	Full	Not available
World Bible — profiles, notes, search	Full	Full
Notes — create and edit	Full	Full

Collaboration — comments, status	Full	Full
Export	Full	Full

16.3 Platform Priority

Platform	Priority	Notes
iPad (native)	Primary	Full product — the definitive PanelSync experience
Web (Expo for Web)	First-class	Identical to iPad minus drawing canvas depth
Android Tablet	v2	Same Expo codebase — testing and input adaptation
iPhone / Android Phone	v2	Quick capture, notes, reading

17. Explicit Non-Goals

- Finished illustration — PanelSync produces rough previs and reference sketches, not print-ready art
- Full brush engine or professional painting tools — this is not Procreate
- Color management or ICC profile support
- Page layout and print production — InDesign territory
- AI-generated imagery — AI may be used for suggestions and recognition, never for panel art
- General-purpose project management — PanelSync is for creative work, not task tracking

Document status: v2.0 — substantially revised. Reflects full product vision including World hierarchy, Timeline, Chalkboard, Relationship Map, Pacing View, and Continuity Tracker.